
Does Cost-Efficiency Travel?

What Agent Leaderboards Can Predict About Cost Across Workloads

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Abstract

1 Agent leaderboards increasingly report task success and dollar cost, inviting users to choose
2 a “cost-efficient” system from a single benchmark. We ask whether that recommendation
3 travels to a new workload. Using public Holistic Agent Leaderboard (HAL) results, we hold
4 the displayed HAL Generalist Agent scaffold fixed and compare repeated displayed model
5 labels across six workloads. Raw dollar-cost ranks are strongly associated in several high-
6 overlap pairs, but this signal is largely absorbed by stable displayed-label and benchmark
7 cost propensity. Held-out prediction is strong for four workloads and uninformative for
8 two lower-overlap workloads. Aggregate dollars per expected success are less stable: only
9 three CostPerSuccess associations survive permutation testing and Holm correction, and
10 the clearest raw-cost versus CostPerSuccess gap occurs for GAIA-SWE-mini. A separate
11 matched public trace study gives the system-level context that leaderboard aggregates lack:
12 across 977 exact task–model pairs, OpenHands uses about 15% fewer median turns and
13 tool calls than SWE-agent, without a detected difference in resolved status. Taken together,
14 the results show that raw cost is a useful but incomplete portability signal. It combines
15 persistent label-level cost propensity with scaffold execution structure, and it should not be
16 treated as a portable ranking of deployment efficiency.

17 1 The practical problem

18 Agent evaluation is expensive enough that cost has become a first-class leaderboard number. HAL reports 21,730
19 rollouts across nine benchmarks and roughly \$40,000 in evaluation cost [Kapoor et al., 2026]. That makes a
20 cost–accuracy frontier appealing: if two systems perform similarly, why not choose the cheaper one?

21 The hidden step in that reasoning is portability. A user sees that a label is inexpensive on one benchmark
22 and assumes it will remain a good buy in a different environment. That assumption is plausible, but it is not
23 innocuous. A reported dollar cost can reflect the model or provider, reasoning mode, prompt and tool setup,
24 cache behavior, scaffold, and the workload itself.

25 The GAIA-SWE-mini comparison makes the issue concrete. Across 17 shared displayed labels, raw dollar-cost
26 ranks are closely aligned (Spearman $\rho = 0.89$). But the ranking by aggregate dollars per expected success is
27 much less aligned ($\rho = 0.29$); the paired raw-minus-CPS contrast is 0.60 with a 95% configuration-resampling
28 sensitivity interval of [0.14, 1.13]. In practical terms, a GAIA-like leaderboard can offer a useful prior about raw
29 spend for a SWE-like workload without providing a reliable ordering of expected dollars per successful task.
30 This is an aggregate ranking example, not a claim about a named model or a production deployment.

31 We therefore ask three questions. First, when does an observed cost ranking predict a held-out workload?
32 Second, does that ranking still travel after separating stable displayed-label cost propensity from benchmark cost
33 level? Third, what can public traces say about the scaffold-level execution structure that aggregate leaderboards
34 hide?

2 What we test, and what the data permit

Our primary data are the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The coverage determines the design. Most scaffolds appear on one workload only, so the data do not support a general model–scaffold–workload decomposition. We instead fix the displayed HAL Generalist Agent scaffold, match exact public model-label strings, and retain six workloads with enough repeated labels: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, and TAU-bench Airline. This cohort contains 86 rows. USACO has one Generalist observation and is excluded.

The primary CSV supplies accuracy and aggregate dollar cost, but not token components, provider aliases, API calls, tool calls, retries, cache usage, or per-task trajectories. Accordingly, we do not call any adjustment “price controlled,” and we do not infer token or failure costs from dollars. Most evaluations are single-run, so rollout-level uncertainty is unavailable.

For each benchmark pair with at least five shared labels, we compute Spearman ρ and Kendall τ_b . Pairwise sensitivity intervals use 10,000 configuration resamples. Raw-cost Holm correction applies to 15 tests. CostPerSuccess uses

$$\text{CPS}_{i,b}(\epsilon) = \frac{C_{i,b}}{\max(A_{i,b}, \epsilon)},$$

where A is accuracy as a fraction and the primary floor is $\epsilon = 0.01$. CPS has separate 20,000-draw label-pairing permutation tests and Holm correction over 15 tests; 5% and zero-exclusion versions are reported in the supplement. CPS is an aggregate dollars-per-expected-success proxy, not the mean cost of successful runs.

To test out-of-sample portability, we use leave-one-benchmark-out (LOBO) prediction. For each held-out workload, we estimate each label’s centered log-cost propensity from the other workloads and predict the held-out cost rank. We also fit the descriptive pooled model

$$\log C_{i,b} = \mu + \alpha_i + \beta_b + \epsilon_{i,b}.$$

This 86-row model has 30 parameters and 56 residual degrees of freedom. Its fit is descriptive; LOBO is the stronger test of portability.

Finally, we report three decision frontiers: discrete nondominance, a HAL-inspired origin-anchored convex hull, and a 5% near-tie tolerance. These answer related but different decision questions.

3 Evidence

3.1 Raw cost predicts some held-out workloads

The most direct evidence comes from LOBO. A label’s cost propensity learned from the other workloads predicts held-out raw cost ranks for CORE ($\rho = 0.78$, $N = 14$, $p = 0.0016$), GAIA (0.87 , $N = 16$, $p < 0.0001$), SWE-mini (0.90 , $N = 16$, $p < 0.0001$), and TAU (0.83 , $N = 14$, $p = 0.0004$). These are not one-label artifacts: leave-one-label-out ranges remain positive for every omission, from $[0.73, 0.91]$ for CORE to $[0.88, 0.92]$ for SWE-mini.

The picture is not uniform. SciCode is unstable under leave-one-label-out analysis ($[-0.50, 0.14]$). ScienceAgentBench remains directionally negative ($[-0.77, -0.43]$), but it has seven labels and its original permutation result is not significant. SciCode has very low mean accuracy (2.1%) and one-third zero-accuracy labels; ScienceAgentBench has the narrowest log-cost dispersion. These are useful diagnostic facts, not a basis for a domain-level causal story.

3.2 What the raw-cost signal appears to contain

Among high-overlap pairs, raw dollar-cost correlations range from 0.74 to 0.93. A pooled benchmark-plus-label log-cost model has $R^2 = 82.0\%$ and adjusted $R^2 = 72.7\%$, but this in-sample fit should not be overread given 30 parameters and 86 rows. Its purpose is diagnostic. Once observed benchmark and displayed-label cost propensity are removed, none of the high-overlap residual rank associations survives a 20,000-draw permutation test.

The reasonable interpretation is not that cost is meaningless. Rather, much of the portable raw-cost signal is a stable observed label-level component. The public data cannot identify whether that component comes from provider pricing, reasoning settings, routing, token behavior, cache treatment, or other unrecorded setup choices.

3.3 Cheapness is not the same ranking as dollars per expected success

CPS changes the question from “which label is cheap?” to “which label buys expected success cheaply?” Under the 1% floor, CPS association survives 20,000-draw permutation testing and Holm correction for CORE–GAIA

84 ($\rho = 0.87$, 95% sensitivity interval [0.63, 0.96]), CORE-TAU (0.83, [0.31, 1.00]), and GAIA-TAU (0.78, [0.42,
85 0.94]). Most other CPS associations remain uncertain.

86 The high-overlap mean CPS correlation is 0.59, compared with 0.84 for raw cost, but that average is descriptive
87 rather than a population-level contrast. The clearest paired difference is GAIA-SWE-mini: raw cost exceeds
88 CPS correlation by 0.60, with a 95% resampling interval [0.14, 1.13]. In short, a raw-spend ranking can travel
89 while a success-adjusted ranking does not.

90 3.4 There is no universal cost-accuracy winner

91 No broadly tested label lies on every frontier under nondominance, the HAL-inspired hull, or the 5% tolerance
92 rule. This is a practical guardrail, not the paper’s central novelty. The stronger statement that no label appears on
93 more than 4/6 frontiers fails under tolerance: o4-mini Low appears on 5/6.

94 4 A separate matched trace study

95 The leaderboard analysis tells us what travels, but not what happens inside an agent run. HAL public trace
96 archives are encrypted. We therefore use a separate public Open-SWE-Traces boundary sample: four documented
97 MiniMax-M2.5 shards under SWE-agent and OpenHands. It is not a random sample of the full corpus. We
98 exactly match task ID and model, exclude duplicate task/model/scaffold records, and obtain 977 pairs.

99 The traces expose resolved status, structured messages, tool calls, and text. They do not expose billing, tokens,
100 provider usage, retries, or latency. We therefore call turns, tool calls, and message characters trace-burden
101 proxies, not cost.

102 At fixed task and model, OpenHands/SWE-agent has a median ratio of 0.85 for trajectory turns (10,000-bootstrap
103 interval [0.82, 0.87]) and 0.85 for tool calls ([0.82, 0.86]). OpenHands has lower values on 69.7% of pairs for
104 each measure. Total trajectory characters have a smaller ratio effect, 0.95 ([0.92, 0.98]), with OpenHands lower
105 on 55.1% of pairs. Resolved status has no detected paired difference ($p = 0.197$). Thus, scaffolds can change
106 visible execution burden even when task and model are held fixed. This does not show lower token use or dollar
107 cost, and it does not identify the mechanism behind HAL rankings.

108 5 Interpretation and implications

109 The evidence supports a narrower conclusion than a single leaderboard recommendation suggests. Raw dollar
110 cost can be predictive across some workloads. In the HAL cohort, that prediction is robust for four held-out
111 workloads and is not driven by a single displayed label. But the signal is partly stable label-level propensity, not
112 a demonstrated measure of portable execution efficiency. Success adjustment changes the ranking and is only
113 partly portable. The matched trace study supplies a compatible systems-level explanation: scaffold choices alter
114 observable work even when task and model are fixed.

115 For practitioners, this means that a leaderboard’s dollar figure is useful as a first screen, not as a final procurement
116 decision. A target evaluation should include a success-adjusted metric and, where possible, trajectory telemetry.
117 For benchmark builders, the missing fields are concrete: model/provider aliases, input/output/cache usage,
118 calls, tool activity, retries, outcome labels, and per-task costs would make it possible to distinguish price from
119 execution behavior.

120 6 Limits

121 This is a secondary-data analysis. Public display labels do not fully identify prompts, tools, budgets, harness
122 versions, or routing. The HAL cohort is sparse and unbalanced, with mostly single-run rows. The public
123 HAL snapshot cannot identify failure-cost ratios, retry behavior, token trajectories, tail cost, or a causal model-
124 scaffold-workload decomposition. CPS is not task-level successful-run cost. The Open-SWE extension is
125 external to HAL and uses a reproducible four-shard boundary sample, not the full corpus. Its proxies are not
126 tokens or dollars. These limits define the claims rather than invalidate them.

127 7 Conclusion

128 A cost-aware leaderboard is better than an accuracy-only leaderboard, but it is not a portable deployment-
129 efficiency ranking by default. In these public agent data, raw cost can generalize to some held-out workloads,
130 yet it partly reflects stable displayed-label propensity and does not always preserve the ranking by dollars per
131 expected success. Matched traces show that scaffold design changes observable execution burden at fixed task

132 and model. The next generation of agent leaderboards should make these distinctions measurable rather than
133 leaving them hidden inside one dollar number.

134 References

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